AIMI: Agentic In-Hand Manipulation for Language-Guided Visual Exploration

Anonymous CVPR submission



Figure 1. **Language-guided visual exploration with AIMI.** The agent chooses how to inspect an object, uses new views to identify the requested face, and guides its presentation. Staged simulation views illustrate the task; recorded trials are reported separately.

Abstract

Dexterous in-hand manipulation is an important skill for robots to reposition and use objects. However, a target specified by language may initially be hidden or blocked by the fingers, requiring the robot to explore before presenting it. In this work, we present AIMI, an agentic in-hand manipulation framework for language-guided visual exploration. The agent chooses inspection actions, identifies the target from new images, and selects a motor goal. Local controllers adjust the motor policy's commands during reasoning and execution. We evaluate our framework in simulation on six objects, each with its own motor policy. Under changed target-to-skill mappings, it achieves 65.8% mean success, exceeding the strongest evaluated baseline by 40.4 percentage points. We demonstrate sim-to-real transfer on four real-world objects using RGB pose feedback, achieving 87.5% success over 80 trials, compared with 66.25% for the strongest evaluated baseline.

1. Introduction

Dexterous in-hand manipulation allows robots to reposition and inspect objects without releasing their grasp. Following language, however, may require finding a surface that the camera cannot see. To "present the face marked T," the robot must expose the mark from behind the object or fingers, identify it, and reach the desired orientation without dropping the object (Fig. 1).

Learning-based controllers reach specified poses or goal images [2, 3, 11]. This provides the motor skills needed for presentation, but goal selection remains difficult when the target is hidden. An instruction specifies the requested surface; visual evidence is needed to select the skill that will present it.

Language-based agents select skills and revise plans from feedback [8, 10]. Within a grasp, this raises two coupled questions: which view should the hand obtain next, and how should it control the object while the agent reasons? Multiple observations may be needed before choosing a presentation goal.

Our goal is language-guided visual exploration with continuous in-hand control. We use hand motion to obtain

evidence for goal selection: an agent chooses inspection actions, examines new images, and links the identified target to a motor goal. This *target binding* guides subsequent presentation.

We implement this approach in AIMI (**A**gentic **I**n-hand **M**anipulation for **I**nspection). We adapt the ViserDex teacher–student foundation to Revo3 [3] and use teacher supervision to connect the student to an agent-selectable goal bank. Local controllers adjust its joint commands during reasoning and execution. While a reply is pending, learned slowing and joint correction keep control responsive to measured feedback.

We evaluate six object-specific simulation systems, achieving 65.8% mean success under changed target-to-skill mappings. Controlled studies examine visual feedback and waiting control, while multilingual trials test instruction coverage. We also demonstrate transfer to four physical objects using RGB pose feedback, with 87.5% success over 80 trials. Our contributions are:

- An agentic framework that connects inspection, target binding and presentation through successive observations for language-guided in-hand visual exploration.
- A goal-conditioned control interface with teacher-supervised local controllers that combine slowing and joint correction during agent reasoning.
- An evaluation across objects, target mappings and languages, with controlled studies of fresh visual evidence and local control, and four-object real-hand validation.

2. Related Work

Agentic reasoning and embodied feedback. Language-based agents connect task descriptions to robot actions. SayCan grounds skill selection in robotic affordances [10], and Inner Monologue uses feedback to update plans [8]. Code as Policies generates programs over perception and control APIs [14], while VoxPoser and ReKep express goals through spatial value maps and keypoint constraints [7, 9]. Magma and VERA further connect multimodal or video models to robot actions [13, 20]. Model inference also takes time while the robot continues moving. Real-time chunking [4], online action correction [18], and delay-aware preference learning [23] address aspects of this problem. LiMA separates sparse intent generation from fast dexterous execution [5]. We study these coupled observation and control needs within an ongoing grasp, where the agent must obtain new views to identify a language-specified target.

Visual and goal-conditioned manipulation. Visual feedback connects object observations to dexterous actions. ViserDex uses Gaussian-splat training images for monocular pose estimation and teacher–student learning for control [3]. GoalDex learns policies conditioned on current and goal images, and uses a VLM to select visual goals from language [2]. Our focus is the exploration needed to select a goal when the requested surface is initially hidden or occluded. New observations guide target binding, while RGB pose feedback supports physical-hand control.

RL for dexterous in-hand manipulation. Simulation-based RL has enabled physical-hand manipulation through domain randomization [1], learning across object shapes [6], and online motor adaptation [15]. RotateIt adds vision and touch [16], and AnyRotate supports varied rotation axes and hand orientations [21]. SimToolReal trains a shared goal-reaching policy on procedural objects for transfer to tool use [11]. Residual learning also supports dexterous bimanual transfer in ManipTrans [12], while latency-aware RL improves generalist policies under inference delays [22]. Our controller study examines teacher-supervised slowing and joint correction around a fixed motor policy while an agent request is pending.

3. Method

AIMI uses in-hand motion to acquire visual evidence and turn it into a presentation goal. An agent decides how to inspect the object, while a recurrent motor policy and local controllers maintain continuous control. Training proceeds from a ViserDex-based teacher–student foundation adapted to Revo3, to goal-bank adaptation, and then local-controller learning (Fig. 2). We describe the cube implementation below. Other objects use their own students and maintenance controllers; their interfaces are given in the supplement.

3.1. Problem formulation

Given an object held in the hand and an instruction ℓ, the robot must find the requested surface target and present it without dropping the object. The target may initially face away from the camera or be occluded by the fingers. Hand motion therefore serves two purposes: exposing new views and reaching the presentation goal.

At step t, the agent observes an RGB image I_t, the instruction and action history. It selects a skill from a fixed goal bank, represented by g_t. The motor policy uses proprioception and history x_t, measured object pose $\hat{p}_t$ and recurrent state h_t to propose joint targets u_t. Local controllers convert these proposals into commands a_t. We write $m_t = 1$ while an agent request is pending. The active skill may be an inspection skill before the target is identified; selecting it does not itself declare that the target has been found.

3.2. Learning goal-conditioned motor skills

Teacher–student foundation. We build on the ViserDex teacher–student approach adapted to the Revo3 hand [3]. The pretrained teacher uses privileged simulation informa-

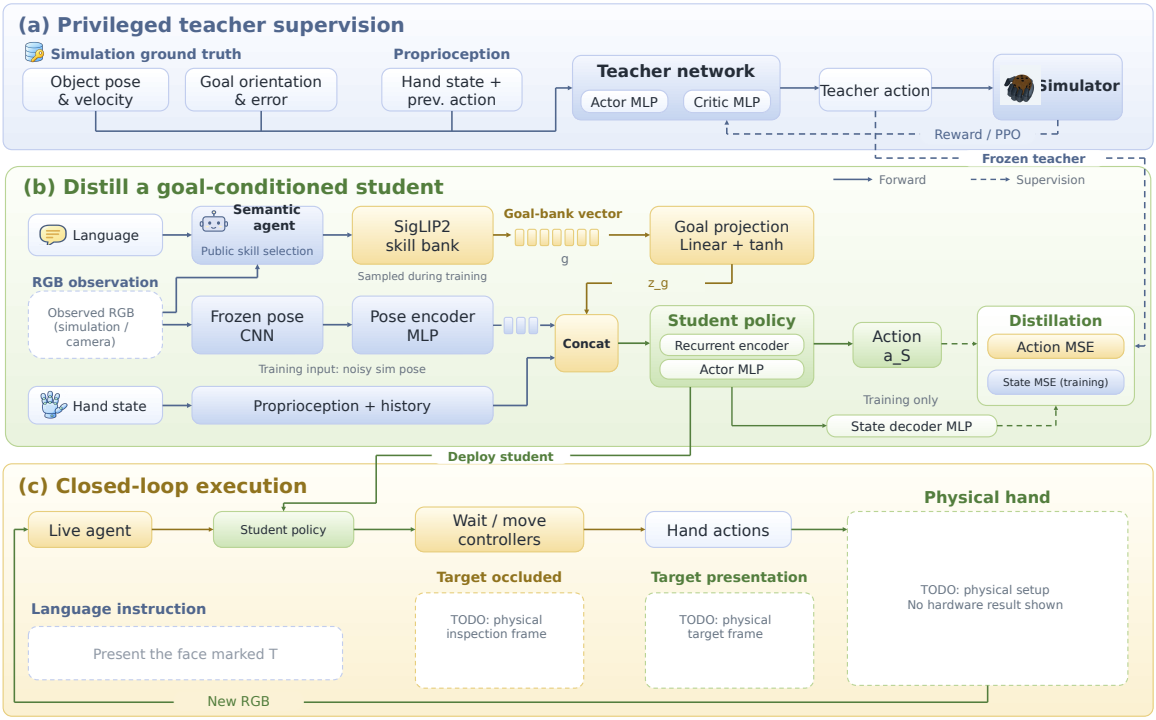


Figure 2. **AIMI training and deployment.** (a) Privileged teacher training follows the ViserDex-based Revo3 foundation. (b) The frozen teacher supervises student adaptation to goal-bank vectors; the state decoder is training-only. (c) An agent selects inspection or presentation skills from new observations while local controllers adjust joint commands. Dashed slots await physical-hand images.

tion and geometric goals to provide action supervision. Starting from the corresponding student, we retain compatible encoders, recurrent and actor weights, and normalization statistics. The teacher remains frozen during the following adaptation.

Goal-bank adaptation. To connect discrete agent decisions to motor skills, we use six normalized vectors $B = \{g_1, \ldots, g_6\}$ from cached SigLIP2 text features [19]. Each represents a color skill and its reference object orientation. A learned projection replaces the student's geometric goal inputs:

$$z_t = \tanh(W_g g_t + b_g), \quad (1)$$

$$(u_t, h_{t+1}) = \pi_\theta([x_t, z_t], \hat{p}_t, h_t). \quad (2)$$

The adapted student receives neither the target quaternion nor the relative-goal quaternion. The goal bank is fixed; online agent outputs select its entries rather than supplying hidden model features.

We fit the projection and recurrent policy to frozen-teacher actions using a DAgger-style procedure [17]. Projection warmup uses action MSE. Recurrent adaptation adds a training-only decoder that reconstructs normalized privileged state with the goal quaternion removed. Its reconstruction MSE has weight 0.2. The teacher and decoder are removed from deployment. Network dimensions and fitting details are provided in the supplement.

3.3. Agentic visual exploration

The agent receives the instruction, current image and action history, and returns an explicit action: inspect, reorient, verify, stop, or abstain. We validate the reply before applying it. Inspection selects a color skill to expose another view. Once the requested face is identified, the agent binds it to a motor goal and guides its presentation (Fig. 3).

The agent chooses inspection and reorientation durations of 10–90 control steps. Later images reveal the effect of these actions and support the next decision. Action duration is distinct from model latency. Only one request can be pending, and the current skill remains active until a valid reply changes it. Invalid replies never select an oracle target. This loop lets the hand gather evidence before committing to a presentation goal, while control continues between decisions.

3.4. Continuous control during and after reasoning

The agent chooses goals at irregular intervals, but the hand needs commands at every control step. We therefore place local controllers around the adapted student. A sensor encoder and a goal encoder feed a fusion layer that predicts

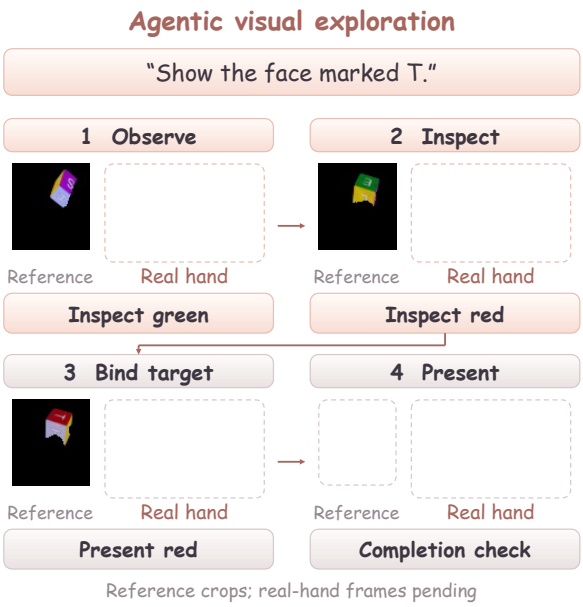


Figure 3. **Agentic visual exploration.** New observations guide inspection, target binding and presentation. Reference crops illustrate the sequence in simulation; physical-hand image slots remain open.

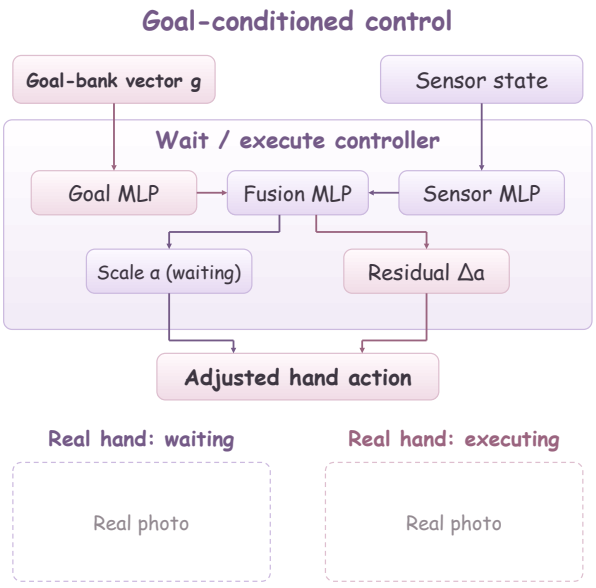


Figure 4. **Continuous local control.** Sensor features and the active goal-bank vector condition joint correction. The waiting controller also adjusts command speed. Empty slots are reserved for physical-hand images.

command corrections (Fig. 4). Sensor features include proprioception, measured pose, the student's recurrent state and proposal, pose-fitting error, and binding and request status.

Waiting and execution commands. The waiting actor also receives the pose and command recorded when the request began, called the *request anchor*, and elapsed waiting time. It predicts a speed logit r_t and joint correction values v_t. With normalized current joints $\bar{q}_t$ and clipped student targets $\bar{u}_t$, the command is

$$\alpha_t = \mathrm{sigmoid}(4r_t + \mathrm{logit}(0.15)), \tag{3}$$

$$b_t = \begin{cases} \bar{q}_t + \alpha_t(\bar{u}_t - \bar{q}_t), & m_t = 1, \\ \bar{u}_t, & m_t = 0, \end{cases} \tag{4}$$

$$a_t = \mathrm{clip}\big(b_t + 0.35\tanh(v_t), -1, 1\big). \tag{5}$$

When no request is pending, a separate execution actor supplies v_t at full proposal speed. Zero waiting-actor outputs recover fixed slowing with coefficient 0.15. Joint correction can also change command direction, which scalar slowing alone cannot do. The underlying joint mapping and smoothing remain unchanged.

Teacher-supervised correction. The frozen teacher supplies holding actions for the request anchor during waiting and actions toward the active goal during execution. We train each actor against these labels with a penalty on its correction $\delta_\phi = 0.35\tanh(v_t)$:

$$\mathcal{L}_{\text{control}} = \mathbb{E}_{(s,g,a^*)\sim\mathcal{D}}\left[\|a_\phi(s,g) - a^*\|_2^2 + 0.01\|\delta_\phi(s,g)\|_2^2\right]. \tag{6}$$

Here s collects sensor features, g is the active goal and a^* is the teacher action. Before target binding, g represents a startup or inspection skill. Deployed actors use measured inputs and the active goal; they cannot access true pose, teacher actions or a pending reply. The execution actor also receives the goal, whose separate benefit is not isolated in our experiments. Controller fitting holds the student fixed, and waiting-actor updates leave the execution actor and stopping rule unchanged.

3.5. RGB perception and closed-loop deployment

For cube control, a frozen CNN takes a 120×120 RGB crop with hand occlusion and predicts image coordinates and depth for the object center and eight keypoints. Camera calibration and rigid fitting recover position, XYZW orientation and a fitting error. Following ViserDex, Gaussian-splat rendering supplies training images for this estimator [3]. The deployed model estimates pose from camera RGB.

At every control step, we update the measured pose, run the recurrent student, choose the waiting or execution actor, and send joint targets to the hand. New agent observations reflect the motion that occurred during inference and execution. All four physical-object systems use this RGB-to-pose-to-

control interface. In simulation, the cube uses RGB pose estimates, while other objects use noisy simulator-derived pose feedback. True pose is reserved for training supervision and independent simulation scoring. Measured-pose stopping and evaluation criteria are specified in Sec. 4.1.

4. Experiments

We organize the evidence around visual exploration (Q3), continuous control during reasoning (Q2), and complete-system performance across objects and instructions (Q1). Q2 and Q3 use the marked cube. Q4 evaluates perception for transfer and complete-system performance on four physical objects.

4.1. Setup and reporting

Objects and systems. Simulation uses Revo3 assets in Isaac Lab 3.0/Newton. Q1 covers a cube, die, duck, hedgehog, pig and milk carton; Q2 and Q3 use the cube. Instructions request one of six surface letters or a dot count on the die. Each target has a fixed reference orientation placing its face upward. Cube systems share the frozen RGB pose estimator, goal bank, student and stopping rule. Other objects use separate students and noisy simulator pose. AIMI additionally uses calibrated completion correction on these objects. The VLM baselines omit the extra control actors and use a fixed inspection schedule when the target is unreadable. These are complete-system differences; Q2 varies only control during pending requests.

Timing and completion. Cube trials allow up to six agent calls over 120 seconds of simulated control at 30 Hz; other objects use 20 Hz. Replies cannot take effect before inference completes, and physics continues during model latency. Simulation runs slower than real time, so these rates are not hardware-speed measurements. The cube stopping check runs at 3 Hz and requires target binding, no pending request or inspection, and measured orientation error at most 0.16 radians. It may terminate without an explicit agent stop action. Independent scoring requires the correct binding, true full-quaternion error at most 0.16 radians at stop, and no drop. There is no extra holding period. Non-cube methods use the same scoring tolerance with object-specific stopping rules. Exact timing, drop boundaries and scoring equations are in the supplement; Q4 gives the physical-hand criterion.

Trials and reporting. Unless specified otherwise, each simulated condition has 40 trials, 20 per evaluation seed, matched by initial physical state and scored target. Q2 uses three seeds with 40 trials each and reports all 120 outcomes per method. Q1 and Q3 use common service-eligible trials: a service error or call exceeding 120 seconds excludes the matched trial from every method. All raw outcomes are retained without replacement. Q1's complete-system trials differ from Q2's controller cohort, so their absolute rates need not match. The supplement provides study-specific seeds, pairing details and statistical tests.

4.2. Q3: Visual Exploration and Feedback

Three paired experiments examine how the agent acquires and uses visual evidence (Table 1): continuing to observe and decide, receiving new images, and choosing inspection direction and duration. Each comparison has its own trial set.

Condition	Success	%	Drops	Calls
(a) Continued feedback: one vs. up to six calls				
One initial plan	6/39	15.4	0	40
Full feedback	33/39	84.6	5	120
(b) Fresh visual evidence: same six-call cap				
Current image	32/40	80.0	3	112
Repeated first frame	9/40	22.5	4	98
(c) Inspection policy: same model and controllers				
Fixed sweep	31/40	77.5	1	126
Adaptive inspection	38/40	95.0	0	117

Table 1. **Q3: visual exploration on the cube.** Each panel schedules 40 trials per condition. (a) excludes one service-affected pair and reuses Q2's initial full-feedback subset. (b) changes only the agent's RGB input. (c) changes inspection direction and duration. (b,c) have no exclusions. Drops/calls include all trials; realized inference costs differ.

(a) Does continued feedback help find the target? An initial plan selects up to six bounded actions from one image and measured feedback; continued feedback allows up to six calls with new images and history. Local controllers, stopping and time limits remain fixed. Success rises from 6/39 to 33/39 eligible trials, with all 27 discordant pairs favoring feedback (exact $p = 1.49 \times 10^{-8}$). Feedback resolves 22 of 28 initial-plan abstentions; 32 trials first bind after inspection (Fig. 5). The next comparison holds the call budget fixed.

(b) Does the next image provide useful evidence? Both conditions use the same six-call cap, request rule, live numeric feedback, history and local control. Only RGB changes: the current image versus the exact first frame. Success is 32/40 versus 9/40, with all 23 discordant pairs favoring fresh images (exact $p = 2.38 \times 10^{-7}$). Correct terminal bindings number 34 versus nine. Both first bind correctly on call one in the same nine trials; fresh images enable 25 additional correct bindings, 23 leading to success, versus none with repeated frames. Drops differ by one; initial visibility is not independently annotated.

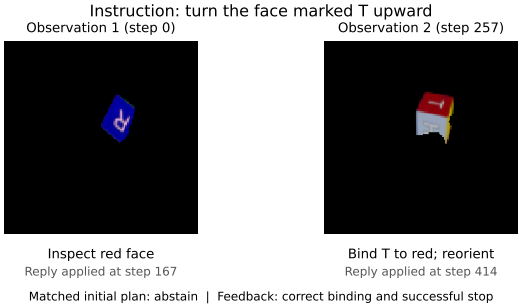


Figure 5. **New evidence enables target binding.** Unretouched model-input frames from a simulated trial. The initial plan abstains; continued feedback exposes T, after which AIMI binds T to red and completes at step 420 (true error 0.0812 radians). Both arms start from the same state and image. This selected example illustrates inspection before binding.

(c) Does choosing the next view help? Forty new pairs compare agent-selected directions and durations with a fixed 90-step color sweep. Both use the same model, controllers, fresh images, history format and stopping; history records executed actions. The agent still decides whether to inspect, bind or abstain. Adaptive inspection achieves 38/40 successes (95.0%) versus 31/40 (77.5%). Nine pairs favor adaptive and two favor fixed inspection: a 17.5-point observed gain (exact $p = 0.0654$). Correct terminal bindings are 39 versus 32, with zero versus one drop. This trend remains statistically unresolved and covers one fixed sweep.

4.3. Q2: Control During Agent Reasoning

Figure 6 compares four ways to control the hand while the agent is reasoning: continue the student's command, slow it by a fixed coefficient of 0.15, learn the slowing coefficient, or learn slowing together with a bounded joint correction. All four use the same execution controller when no request is pending. Each method has 120 trials: 40 under each of three evaluation seeds, with the trained weights fixed. These seeds vary evaluation tasks, not training initialization.

Learned slowing with joint correction achieves 104/120 successes (86.7%), compared with 88/120 (73.3%) for fixed slowing, a gain of 13.3 percentage points. Waiting drops decrease from 13 to five, and execution drops from eight to four. Learning speed alone gives 82/120 successes (68.3%), below fixed slowing. The results favor learning speed and joint correction together. Drop stages tell us when failure occurs, but do not identify its cause. Seed settings and outcome breakdowns are in the supplement.

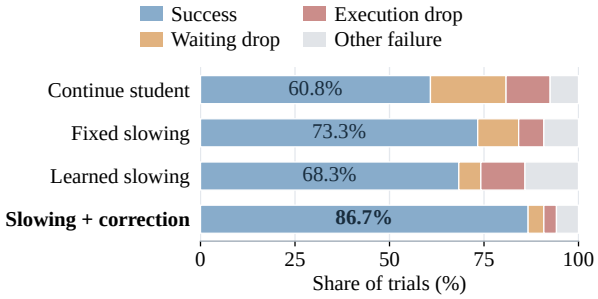


Figure 6. **Q2: outcomes during controller ablation.** Each bar contains 120 trials, 40 per evaluation seed. Labels show success rates; colored segments partition all outcomes. Drop stages record when failure occurs. Full counts are in the supplement.

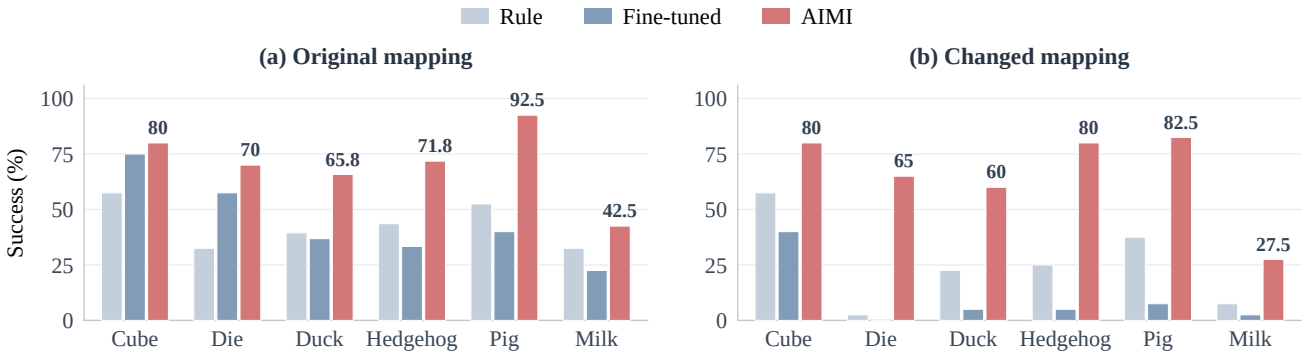


Figure 7. **Q1: task success across six simulated objects.** Bars use common service-eligible trials: 40 per cell, except 38 for original-mapping duck and 39 for original-mapping hedgehog. Labels show AIMI rates. Each object uses its own policy. The supplement retains all values and system differences. These are complete-system comparisons.

4.4. Q1: Objects, Target Mappings and Language

We compare original and changed target-to-skill mappings on six objects (Fig. 7), then test language and wording coverage. Separate native-image cohorts and their inconclusive AIMI–fine-tuned comparisons are reported in the supplement.

Aggregate system performance. Across 1,440 scheduled trials, mean success is 70.4% with original mappings and 65.8% with changed mappings, exceeding the strongest baseline means (44.2% and 25.4%) by 26.2 and 40.4 points. These descriptive, post-hoc means weight each object's service-eligible rate equally.

Four additional objects. Duck, hedgehog, pig and milk carton contribute 960 trials, with AIMI leading in each condition. Three service failures exclude matching trials from all methods. Milk-carton drops remain frequent: 15/40 original-mapping and 20/40 changed-mapping AIMI trials. Reset checks and sensitivity analyses appear in the supplement.

Visual-semantic intervention. Only the agent's RGB changes across 40 matched states with original, reassigned or hidden letters; pose-estimator input and lighting remain fixed. Reassignment leaves AIMI at 32/40 successes but reduces fine-tuning from 30/40 to 16/40 (Holm $p = 0.0130$, six comparisons); rules remain at 23/40. Hiding letters gives 0/39, 3/40 and 0/40, respectively. Nine of eleven fine-tuned wrong bindings follow the old mapping. One reassignment does not establish broad generalization.

Dice layout transfer. Across 240 die trials, rotating the visual mesh preserves collision geometry, inertia and the skill-ID prompt. Exploratory paired tests favor AIMI over both baselines under the changed layout; its own layout difference remains inconclusive. The supplement reports all seven unadjusted contrasts.

Language and wording coverage. We test standard and rewritten English and Chinese instructions on three objects: 60 trials per condition across three seeds, totaling 2,160 trials (Table 2). Weights, mappings and scoring remain fixed, without Chinese retraining. The rule baseline also translates its task/output instructions. Raw outcomes remain in the supplement.

	English		Chinese	
System	Standard	Rewrite	Standard	Rewrite
Cube (n)	56	59	57	53
Rule VLM	48.2	45.8	31.6	32.1
Fine-tuned VLM	71.4	72.9	75.4	**81.1**
AIMI	**75.0**	**79.7**	**84.2**	67.9
Duck (n)	60	59	60	60
Rule VLM	50.0	25.4	33.3	31.7
Fine-tuned VLM	38.3	30.5	31.7	35.0
AIMI	**68.3**	**67.8**	**68.3**	**76.7**
Hedgehog (n)	60	60	52	60
Rule VLM	46.7	38.3	36.5	36.7
Fine-tuned VLM	38.3	31.7	34.6	25.0
AIMI	**73.3**	**76.7**	**75.0**	**76.7**

Table 2. **Q1: language and wording coverage.** Completion (%) on common service-eligible trials. Methods share the listed n of 60 scheduled slots. Bold denotes the highest observed rate, not significance. Raw counts and paired analyses are in the supplement.

AIMI exceeds fine-tuning in all eight duck/hedgehog conditions and rules in six after Holm correction over 36 method tests. Cube comparisons remain inconclusive, with fine-tuning leading for rewritten Chinese. Strict state/image pairing preserves these decisions. Language/wording contrasts remain inconclusive after a separate 36-test correction.

4.5. Q4: Pose Estimation and Sim-to-Real Transfer

Perception adaptation for transfer. We adapt the ViserDex-based perception pipeline with real-camera calibration and Gaussian-splat appearance assets. Table 3 evaluates RGB and RGB-D variants, showing improved pose recall with RGB-D in simulation. Training and camera adaptation differ between variants. This supporting perception study is separate from the RGB-pose hardware evaluation below.

Estimator	Pose recall (%) ↑		Invalid ↓
	10 mm / 10°	5 mm / 5°	
RGB	37.53	14.63	69
RGB-D	**64.73**	**47.56**	**1**

Table 3. **Q4: independent pose estimation.** Recall requires both errors below the listed thresholds. All 6,144 simulated observations remain in the denominator, including invalid predictions. Evaluation uses ideal masks and virtual depth.

Complete-system real-hand evaluation. We evaluate Rule, Fine-tuned and AIMI on a cube, die, ball and duck, with 20 trials per object and method (240 total). All use RGB-based pose estimation. A trial succeeds when the hand reaches the requested target and stops within 15° of its target orientation without dropping the object. Table 4 reports per-object success counts and rates. Overall rates are 40.0%, 66.25% and 87.5%, respectively. AIMI completes 70/80 trials, exceeding fine-tuning by 21.25 percentage points.

Object	Rule	Fine-tuned	AIMI
Cube	9/20 (45%)	15/20 (75%)	**18/20 (90%)**
Die	8/20 (40%)	14/20 (70%)	**18/20 (90%)**
Ball	9/20 (45%)	13/20 (65%)	**19/20 (95%)**
Duck	6/20 (30%)	11/20 (55%)	**15/20 (75%)**
Total	32/80 (40%)	53/80 (66.25%)	**70/80 (87.5%)**

Table 4. **Q4: real-hand task completion.** Successes/trials and rates for the four physical objects. Every object–method cell contains 20 trials. Success requires reaching the requested target and stopping within 15° without dropping the object.

5. Discussion and Limitations

Exploration and control. Q3 links new visual evidence to target binding and completion; the adaptive-inspection trend remains statistically unresolved. Q2 supports joint correction during reasoning. Q1 compares complete systems differing in models, history, controllers and stopping.

Evaluation limits. Equal call limits do not imply equal inference cost. Controller-goal conditioning remains unisolated, and Q3 tests one fixed sweep on one cube. Exploration is directed toward six predefined surface targets per object. Simulation completion measures binding and terminal pose; independent visibility and post-stop holding require separate reporting.

Deployment scope. Each object uses separate weights. Simulation uses RGB pose for cube and noisy simulator pose otherwise; hardware uses RGB pose. Hardware results do not establish unseen-object transfer or isolate perception gains. Real-camera pose accuracy remains unmeasured. Hosted-model changes limit reproducibility; service-eligible rates do not measure end-to-end availability.

6. Conclusion

AIMI connects language-guided visual exploration to continuous in-hand manipulation. New observations guide target identification while local controllers adjust motion. Cube studies support fresh visual feedback and joint correction during reasoning. Evaluations cover six simulated objects, two languages, and four physical objects with 87.5% success over 80 hardware trials.

LLM Usage Statement

OpenAI Codex assisted with research discussion, implementation, experiment orchestration, literature retrieval, statistical analysis, figure-layout code, and drafting and editing throughout the manuscript. A hosted vision-language model and Qwen3-VL are evaluated system components. Numerical claims are grounded in recorded experiments; unfinished work is identified explicitly. The authors are responsible for verifying the text, references, code, figures, and results before submission.

References

[1] Marcin Andrychowicz, Bowen Baker, Maciek Chociej, Rafal Józefowicz, Bob McGrew, Jakub Pachocki, Arthur Petron, Matthias Plappert, Glenn Powell, Alex Ray, Jonas Schneider, Szymon Sidor, Josh Tobin, Peter Welinder, Lilian Weng, and Wojciech Zaremba. Learning dexterous in-hand manipulation. *The International Journal of Robotics Research*, 39(1):3–20, 2020. 2

[2] Arjun Bhardwaj, Mayank Mittal, Maximum Wilder-Smith, Vaishakh Patil, and Marco Hutter. GoalDex: Learning image-goal-conditioned policies for dexterous in-hand manipulation. *arXiv preprint arXiv:2611.02171*, 2026. 1, 2

[3] Arjun Bhardwaj, Maximum Wilder-Smith, Mayank Mittal, Vaishakh Patil, and Marco Hutter. ViserDex: Visual sim-to-real for robust dexterous in-hand reorientation. *arXiv preprint arXiv:2604.11138*, 2026. 1, 2, 4

[4] Kevin Black, Manuel Galliker, and Sergey Levine. Real-time execution of action chunking flow policies. In *Advances in Neural Information Processing Systems*, pages 33383–33407, 2025. 2

[5] Ning Chen, Yankai Fu, Junkai Zhao, Qianpu Sun, Guocai Yao, Pengwei Wang, Zhongyuan Wang, and Shanghang Zhang. LiMA: Bridging long-term imagination to real-time dexterous manipulation via asynchronous diffusion. *arXiv preprint arXiv:2609.28431*, 2026. 2

[6] Tao Chen, Jie Xu, and Pulkit Agrawal. A system for general in-hand object re-orientation. In *Proceedings of the 5th Conference on Robot Learning*, pages 297–307, 2022. 2

[7] Wenlong Huang, Chen Wang, Ruohan Zhang, Yunzhu Li, Jiajun Wu, and Li Fei-Fei. VoxPoser: Composable 3D value maps for robotic manipulation with language models. In *Proceedings of the 7th Conference on Robot Learning*, pages 540–562, 2023. 2

[8] Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, Pierre Sermanet, Tomas Jackson, Noah Brown, Linda Luu, Sergey Levine, Karol Hausman, and Brian Ichter. Inner monologue: Embodied reasoning through planning with language models. In *Proceedings of the 6th Conference on Robot Learning*, pages 1769–1782, 2023. 1, 2

[9] Wenlong Huang, Chen Wang, Yunzhu Li, Ruohan Zhang, and Li Fei-Fei. ReKep: Spatio-temporal reasoning of relational keypoint constraints for robotic manipulation. In *Proceedings of the 8th Conference on Robot Learning*, pages 4573–4602, 2025. 2

[10] Brian Ichter, Anthony Brohan, Yevgen Chebotar, Chelsea Finn, Karol Hausman, Alexander Herzog, Daniel Ho, Julian Ibarz, Alex Irpan, Eric Jang, Ryan Julian, Dmitry Kalashnikov, Sergey Levine, Yao Lu, Carolina Parada, Kanishka Rao, Pierre Sermanet, Alexander T. Toshev, Vincent Vanhoucke, Fei Xia, Ted Xiao, Peng Xu, Mengyuan Yan, Noah Brown, Michael Ahn, Omar Cortes, Nicolas Sievers, Clayton Tan, Sichun Xu, Diego Reyes, Jarek Rettinghouse, Jornell Quiambao, Peter Pastor, Linda Luu, Kuang-Huei Lee, Yuheng Kuang, Sally Jesmonth, Nikhil J. Joshi, Kyle Jeffrey, Rosario Jauregui Ruano, Jasmine Hsu, Keerthana Gopalakrishnan, Byron David, Andy Zeng, and Chuyuan Kelly Fu. Do as i can, not as i say: Grounding language in robotic affordances. In *Proceedings of the 6th Conference on Robot Learning*, pages 287–318, 2023. 1, 2

[11] Kushal Kedia, Tyler Ga Wei Lum, Jeannette Bohg, and C. Karen Liu. SimToolReal: An object-centric policy for zero-shot dexterous tool manipulation. *arXiv preprint arXiv:2602.16863*, 2026. 1, 2

[12] Kailin Li, Puhao Li, Tengyu Liu, Yuyang Li, and Siyuan Huang. ManipTrans: Efficient dexterous bimanual manipulation transfer via residual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6991–7003, 2025. 2

[13] Sizhe Lester Li, Evan Kim, Xingjian Bai, Tong Zhao, Tao Pang, Max Simchowitz, and Vincent Sitzmann. Turning video models into generalist robot policies. *arXiv preprint arXiv:2605.27817*, 2026. 2

[14] Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. Code as policies: Language model programs for embodied control. In *Proceedings of the IEEE International Conference on Robotics and Automation*, pages 9493–9500, 2023. 2

[15] Haozhi Qi, Ashish Kumar, Roberto Calandra, Yi Ma, and Jitendra Malik. In-hand object rotation via rapid motor adaptation. In *Proceedings of the 6th Conference on Robot Learning*, pages 1722–1732, 2023. 2

[16] Haozhi Qi, Brent Yi, Sudharshan Suresh, Mike Lambeta, Yi Ma, Roberto Calandra, and Jitendra Malik. General in-hand object rotation with vision and touch. In *Proceedings of the 7th Conference on Robot Learning*, pages 2549–2564, 2023. 2

[17] Stephane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, pages 627–635, 2011. 3

[18] Kohei Sendai, Maxime Alvarez, Tatsuya Matsushima, Yutaka Matsuo, and Yusuke Iwasawa. Leave no observation behind: Real-time correction for VLA action chunks. *arXiv preprint arXiv:2509.23224*, 2025. 2

[19] Michael Tschannen, Alexey Gritsenko, Xiao Wang, Muhammad Ferjad Naeem, Ibrahim Alabdulmohsin, Nikhil Parthasarathy, Talfan Evans, Lucas Beyer, Ye Xia, Basil Mustafa, Olivier Hénaff, Jeremiah Harmsen, Andreas Steiner, and Xiaohua Zhai. SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization,

and dense features. *arXiv preprint arXiv:2502.14786*, 2025. 3

[20] Jianwei Yang, Reuben Tan, Qianhui Wu, Ruijie Zheng, Baolin Peng, Yongyuan Liang, Yu Gu, Mu Cai, Seonghyeon Ye, Joel Jang, Yuquan Deng, and Jianfeng Gao. Magma: A foundation model for multimodal AI agents. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14203–14214, 2025. 2

[21] Max Yang, Chenghua Lu, Alex Church, Yijiong Lin, Christopher J. Ford, Haoran Li, Efi Psomopoulou, David A. W. Barton, and Nathan F. Lepora. AnyRotate: Gravity-invariant in-hand object rotation with sim-to-real touch. In *Proceedings of the 8th Conference on Robot Learning*, pages 4727–4747, 2025. 2

[22] Brian Zhu, Momen Khalil, E Harrison, Emanuele Poggi, Philipp Schmitt, Bernd Kast, Philine Meister, Pranav Atreya, Qiyang Li, Finn Ferchau, Cesar Colmenero, Yash Shahapurkar, Gokul Narayanan, Melih Erdogan, Kai Wurm, Georg von Wichert, Oier Mees, Eugen Solowjow, Andrew Wagenmaker, and Sergey Levine. Learning to act while waiting: RL finetuning of generalist robot policies under inference latency. *arXiv preprint arXiv:2608.23831*, 2026. 2

[23] Yixiang Zhu, Yonghao Chen, Jiaxiang Zou, Yusong Hu, and Xinyu Chen. DEFLECT: Temporal counterfactual preference learning for delay-robust asynchronous VLAs. *arXiv preprint arXiv:2605.19294*, 2026. 2